DATASET ANALYSIS · VERSION 1.0 · OCTOBER 2026

# What the data *says today*

Every dataset the project holds, measured: how the outcome tables are built, whether they agree with themselves, what the length-of-stay and discharge figures show, how the acuity instrument is structured, and exactly which questions these data can and cannot answer.


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Data analysis, figures, reproducible notebook and reporting.


**SOURCE**
Center 1, 2016–2026, aggregated — no patient-level records

**REPRODUCED BY**
`notebooks/01_dataset_analysis.ipynb`

**SYNTHETIC DATA**
None. Every figure comes from a file in `datasets/`

**DECISION SUPPORTED**
What can be modelled now, and what must be requested

# 00 Contents, and the finding in brief

One page for a busy reader; the detail follows.


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**The finding in three sentences.** The project holds two real bodies of data: a decade of one centre's aggregated neonatal outcomes, and the unit's acuity instrument transcribed into tables. They are internally consistent and strong enough to describe how length of stay and discharge outcome vary with birth weight and gestation — which is what the application does today. They cannot support a fitted model, a prediction for an individual infant, or any statement about staffing over time, because they contain no individual records, no dates and no workforce data.

| 7 | 712 | 127 | 0 |
|---|---|---|---|
| DATA FILES PARSED | ADMISSIONS BEHIND THE FIGURES | PUBLISHED STATISTICS | SYNTHETIC ROWS |

**Synthetic data has been removed from this project.** An earlier version of NeoStay carried a simulated year of unit activity for demonstration. It has been deleted from the repository, and nothing in this document, in the application or in the notebook is generated, simulated or imputed.

# 01 Everything the project holds

Eighteen files, three folders, two bodies of evidence.

The source of truth is small and fully inspectable. Seven CSV files are read by the application; the remainder are the centre's own charts of the same numbers, the scanned original of the acuity instrument, and two local policy documents.

**TABLE 1 — THE DATASETS, AND WHAT READS THEM**

| File | What it contains | Read by |
|---|---|---|
| **losdata/** … `_by-Birth Wgt 10 Levels-survival.csv` | Six measures per birth-weight band: initial length of stay split by discharge outcome (home, transfer, died, all), a predicted length of stay, and total length of stay. N, median, Q1 and Q3 for each. | `data_loader.py` |
| **losdata/** … `Total Length Of Stay…by-Birth Wgt 10 Levels.csv` | Total length of stay across the same 11 birth-weight bands. | `data_loader.py` |
| **losdata/** … `Total Length Of Stay…by-GA Week.csv` | Total length of stay across 18 gestational-age groups, from under 22 weeks to over 37 weeks. | `data_loader.py` |
| **losdata/** … `by-Birth Wgt 10 Levels.csv` | Predicted and total length of stay by band. Every value duplicates a value in the survival file. | not read — duplicate |
| **losdata/** `*.png`, `*.pdf` (8 files) | The centre's own charts of the tables above, kept for provenance. | not read |
| **nurse-skills/** `page_1…csv`, `page_2…csv`, `page_3…csv` | The Abrazo Arrowhead Neonatal Acuity Tool and the Nursing Levels of Care, transcribed from the scan into tables. | `acuity_tool.py` |
| **nurse-skills/** `Nurse-skills-classifier….pdf` | The original three-page scan, 11 MB. Its handwritten annotations are deliberately ignored. | not read by the program |
| **nurse-poicies/** `Abrazo_Acuity_Tool.docx` | The same acuity tool as a Word document — a second, cleaner copy of the transcription source. | not read |
| **nurse-poicies/** `NICU_Charge_Nurse_Daily_Task_List.docx` | The charge nurse's hour-by-hour duties across the AM and PM shifts. | not read |

## The cohort behind the outcome tables

All four outcome files describe the same population: **very low birth weight infants admitted to Center 1 between 2016 and 2026**, summarised as counts, medians and quartiles. There is no row for an individual infant anywhere, no admission date, and no clinical detail beyond birth weight and gestational age.

**An unused document worth reading.** The charge nurse task list is the only description the project holds of how a shift is actually run. It names the local artefacts a future data request should ask for by name: the *staffing matrix guides* used to make assignments, the *Monthly Acuity Tracker*, the *NICU Denominator sheet*, the handoff report and the first-admit / float / low-census note written at 15:00. Each is a record the unit already produces, which makes it far cheaper to request than a new collection.

# 02 How the outcome tables are built

A wide shape, read into tidy rows, then checked against the application.

Each outcome CSV is a *wide* table. The first row names the groups — birth-weight bands, or gestational-age weeks — and the second row repeats four sub-columns beneath every group. Each later row is one measure.

**TABLE 2 — THE SHAPE OF EVERY OUTCOME FILE**

| Row | Contents |
|---|---|
| Row 1 | Group labels: `< 501 g`, `501-600 g` … `> 1400 g`, `All` — each spanning four columns. |
| Row 2 | `N`, `Median`, `Q1`, `Q3`, repeated once per group. |
| Row 3+ | One measure per row, for example *Total Length Of Stay* or *Initial Length of Stay (By Disposition) - Died*. |

The notebook parses this shape into one tidy row per measure and group — **127 statistics in all** — using its own parser, deliberately independent of the application's loader. It then compares 40 of those rows against `backend/app/data_loader.py`. They agree exactly, so the figures in this document and the numbers the application shows come from the same reading of the same files.

> **Why quartiles matter more than means here.** The files publish medians and quartiles, not means or standard deviations. Length of stay is strongly right-skewed — a few infants stay for most of a year — so the median is the honest centre, and the distance from Q1 to Q3 is the only measure of spread available. Every estimate the application makes carries that interquartile band with it rather than quoting a single number.

# 03 Completeness and internal consistency

Four checks, run over every published statistic.

**TABLE 3 — AUTOMATED CHECKS AND THEIR RESULTS**

| Check | Scope | Result |
|---|---|---|
| Every expected group present | 11 weight bands or 18 gestational-age groups, in every measure of every file | pass no gaps |
| Q1 ≤ median ≤ Q3 | All 127 statistics | pass 0 violations |
| Band counts sum to the `All` column | 10 measure/file combinations | pass 10 of 10 |
| Values repeated across files agree | 24 measure/group pairs published in two files | pass 0 disagreements |

Structurally the tables are clean. The substantive finding is elsewhere: **the measures do not share a denominator.**

**TABLE 4 — HOW MANY INFANTS STAND BEHIND EACH MEASURE**

| Measure | Infants | Difference from the disposition total |
|---|---|---|
| Initial length of stay — discharged home | 553 | 77.7% of all admissions |
| Initial length of stay — transferred | 73 | 10.3% |
| Initial length of stay — died | 83 | 11.7% |
| **Sum of the three outcomes** | 709 | — |
| **Disposition total (the `All` row)** | 712 | 3 admissions carry no discharge outcome |
| Total length of stay | 708 | 4 fewer than the disposition total |
| Predicted length of stay | 618 | 94 fewer — a visibly smaller group |

**What this means for modelling.** Three admissions have no recorded outcome, four have no total stay, and the predicted-stay measure covers 94 fewer infants than the cohort. With aggregated data there is no way to tell *which* infants are missing, or whether they differ from the rest. Rates in this document therefore always name the denominator used, and any future model will need the individual records to resolve the gap rather than assume it is random.

The missing values are also unevenly spread: the three unattributed admissions fall in the `< 501 g`, `501-600 g` and `> 1400 g` bands, one each. That pattern is consistent with infants still admitted when the extract was taken, but the files do not say so, and it is a question worth putting to the centre.

# 04 Length of stay by birth weight

The gradient the application's forecast rests on.

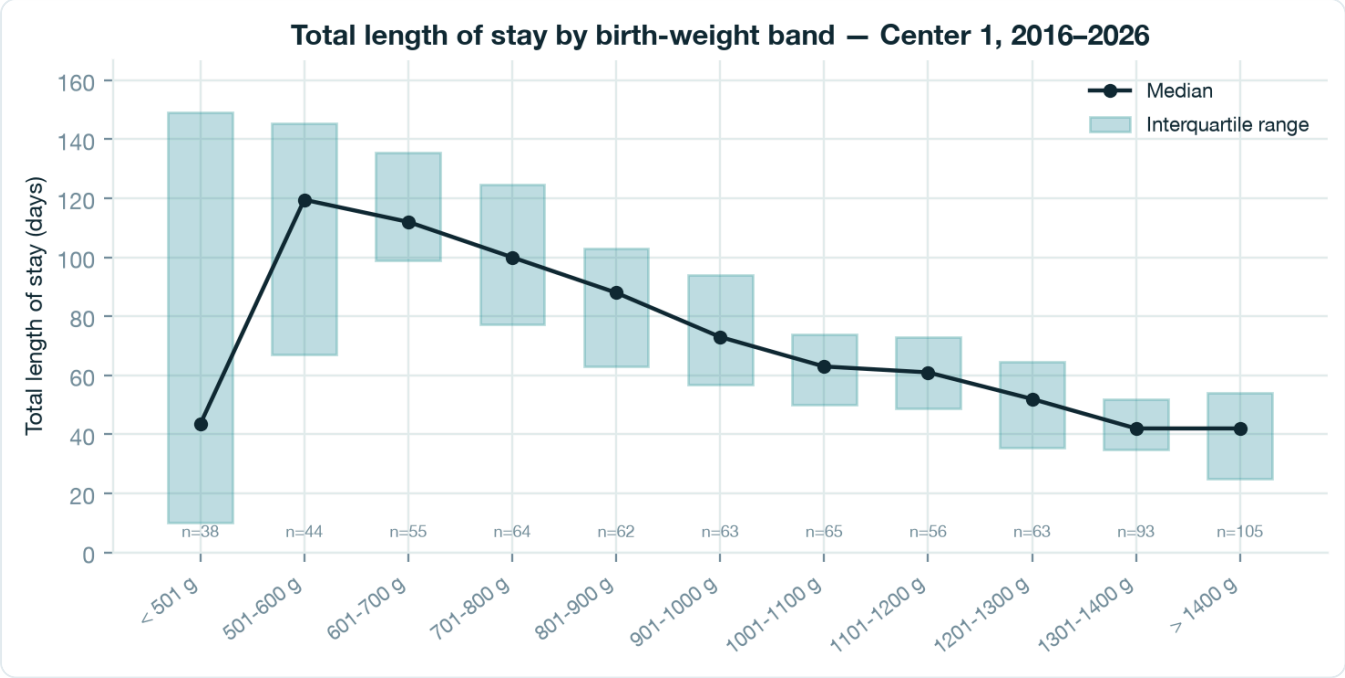



**Figure 1 — Total length of stay by birth-weight band.** The bar spans Q1 to Q3; the line tracks the median; the count behind each band is printed at its foot. Center 1, 2016–2026, 708 infants.

**TABLE 5 — TOTAL LENGTH OF STAY, IN DAYS, BY BIRTH-WEIGHT BAND**

| Band | Infants | Median | Q1 | Q3 | IQR width |
|---|---|---|---|---|---|
| < 501 g | 38 | 43.5 | 10.0 | 149.0 | 139.0 |
| 501–600 g | 44 | 119.5 | 67.0 | 145.3 | 78.3 |
| 601–700 g | 55 | 112.0 | 99.0 | 135.5 | 36.5 |
| 701–800 g | 64 | 100.0 | 77.3 | 124.8 | 47.5 |
| 801–900 g | 62 | 88.0 | 63.0 | 103.0 | 40.0 |
| 901–1000 g | 63 | 73.0 | 57.0 | 94.0 | 37.0 |
| 1001–1100 g | 65 | 63.0 | 50.0 | 74.0 | 24.0 |
| 1101–1200 g | 56 | 61.0 | 48.8 | 73.0 | 24.3 |
| 1201–1300 g | 63 | 52.0 | 35.5 | 64.5 | 29.0 |
| 1301–1400 g | 93 | 42.0 | 35.0 | 52.0 | 17.0 |
| > 1400 g | 105 | 42.0 | 25.0 | 54.0 | 29.0 |
| **All** | **708** | **60.0** | **40.0** | **95.0** | **55.0** |

Three readings matter:

- **The gradient is steep and orderly.** From 501–600 g to over 1400 g the median stay falls from about four months to six weeks, monotonically, with no reversal.
- **Uncertainty shrinks as weight rises.** The interquartile width narrows from 78 days in the 501–600 g band to 17 days at 1301–1400 g: heavier infants are not only quicker to go home, they are more predictable.
- **The smallest band breaks the pattern.** At under 501 g the median is the second *lowest* of all eleven bands, and the interquartile range is the widest at 139 days. Section 5 explains why — and why a model must treat survival and stay together.

# 05 Discharge outcome and survival

Three different processes behind one length of stay.

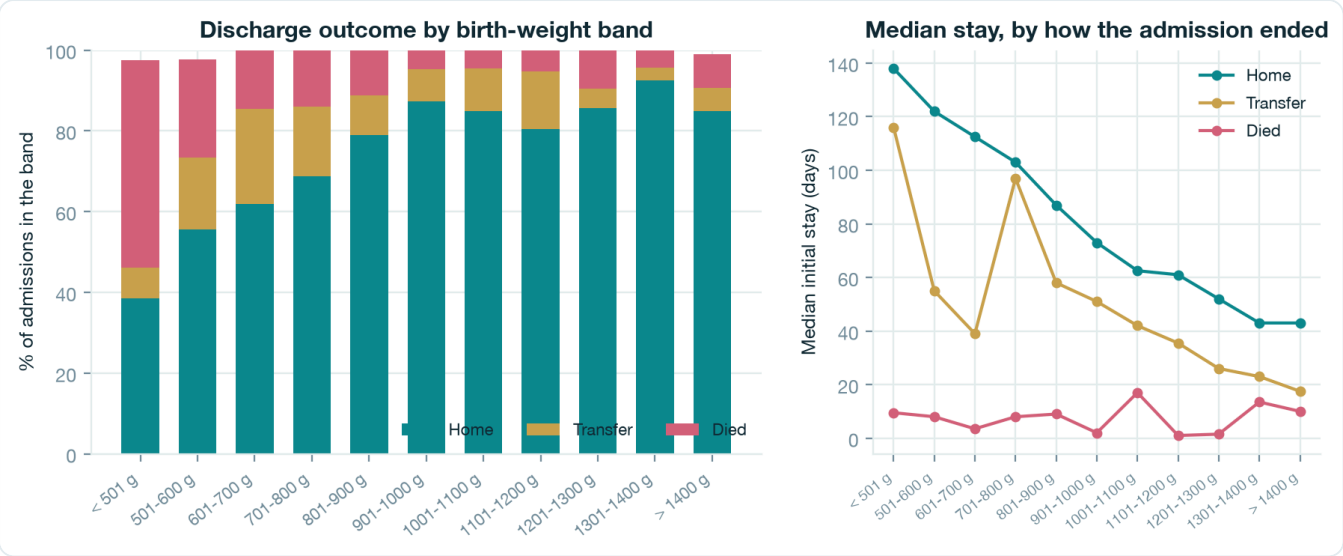



**Figure 2 — How admissions ended, and how long each ending took.** Left: the share of each band discharged home, transferred or died. Right: the median initial stay for each of those three endings.

### TABLE 6 — DISCHARGE OUTCOME BY BIRTH-WEIGHT BAND (COUNTS, AND % OF THE BAND)

| Band | Home | Transfer | Died | All | Home % | Died % |
|---|---|---|---|---|---|---|
| < 501 g | 15 | 3 | 20 | 39 | 38.5 | 51.3 |
| 501–600 g | 25 | 8 | 11 | 45 | 55.6 | 24.4 |
| 601–700 g | 34 | 13 | 8 | 55 | 61.8 | 14.5 |
| 701–800 g | 44 | 11 | 9 | 64 | 68.8 | 14.1 |
| 801–900 g | 49 | 6 | 7 | 62 | 79.0 | 11.3 |
| 901–1000 g | 55 | 5 | 3 | 63 | 87.3 | 4.8 |
| 1001–1100 g | 56 | 7 | 3 | 66 | 84.8 | 4.5 |
| 1101–1200 g | 45 | 8 | 3 | 56 | 80.4 | 5.4 |
| 1201–1300 g | 54 | 3 | 6 | 63 | 85.7 | 9.5 |
| 1301–1400 g | 86 | 3 | 4 | 93 | 92.5 | 4.3 |
| > 1400 g | 90 | 6 | 9 | 106 | 84.9 | 8.5 |
| **All** | **553** | **73** | **83** | **712** | **77.7** | **11.7** |

**Mortality is concentrated at the smallest sizes.** Half of the admissions under 501 g did not survive, against roughly one in twenty from 901 g upward. Because a death usually ends an admission early — the median stay

for infants who died is 6 days, against 63 days for those discharged home — the smallest band's apparently short stay is a survival effect, not a clinical one.

> **Why the application reports both together.** A single length-of-stay figure averages three different processes: going home, being transferred, and dying. NeoStay shows the discharge mix beside every forecast for exactly this reason, and any fitted model will have to treat these as competing risks rather than as one event.

Above 901 g the remaining variation in mortality (4.3% to 9.5%) rests on three to nine deaths per band. Those counts are too small to read as a trend, and the figure should not be used to rank bands against one another in that range.

# 06 Length of stay by gestational age

The same message, with much thinner tails.

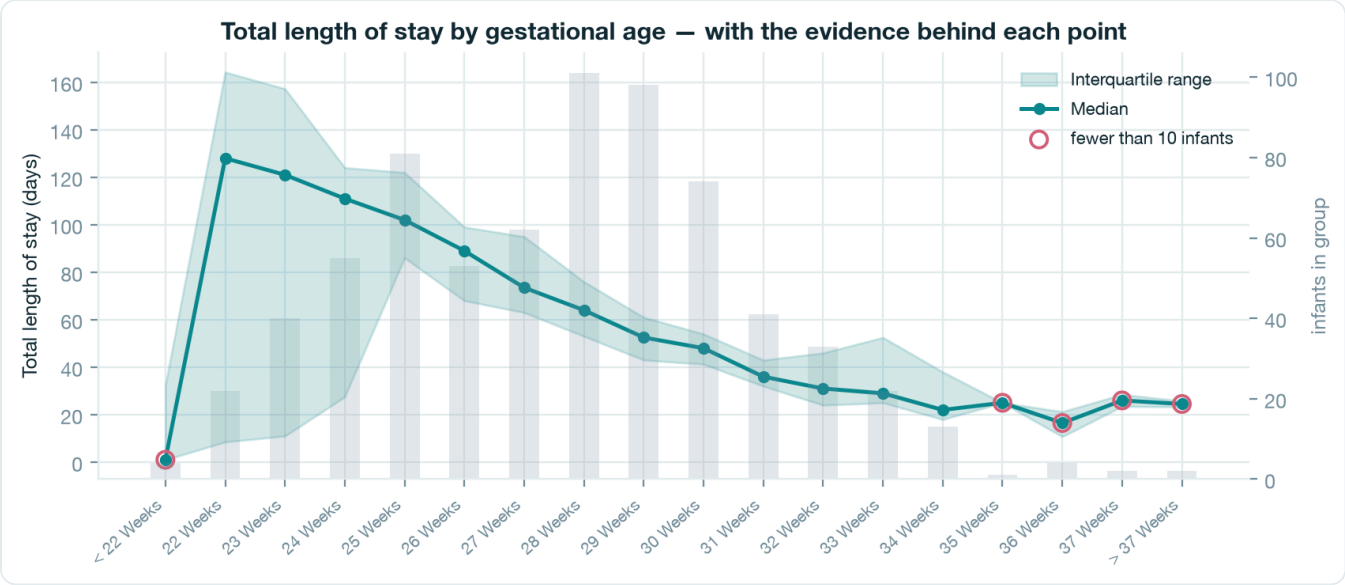



**Figure 3 — Total length of stay by gestational age.** The shaded band is Q1 to Q3; grey bars behind the line show how many infants sit in each group; circled points mark groups with fewer than ten infants.

The gestational-age view carries the same gradient as the weight view — each additional week before birth shortens the expected stay — but its ends are thin. Five groups hold fewer than ten infants:

**TABLE 7 — GROUPS TOO SMALL TO ESTIMATE SEPARATELY**

| Group | Infants | Median stay | Reading |
|---|---|---|---|
| < 22 weeks | 4 | 1.0 | Four infants; almost all die within a day. |
| 35 weeks | 1 | 25.0 | A single infant. No spread can be estimated. |
| 36 weeks | 4 | 16.5 | Growth-restricted infants; very low birth weight at 36 weeks is unusual. |
| 37 weeks | 2 | 26.0 | Two infants. |
| > 37 weeks | 2 | 24.5 | Two infants. |

The evidence sits in the middle of the distribution: 28 weeks (101 infants), 30 weeks (98), 29 weeks (98) and 25 weeks (81). The application's confidence grade is driven by exactly this count, so a thin group is reported as low-confidence rather than hidden.

**A limit worth stating plainly.** The weight table and the gestational-age table describe the *same* 708 infants in two different ways, and no cross-tabulation is published. The joint effect of weight and gestation therefore cannot be measured — only approximated by blending the two margins, which is what NeoStay does and says it does. Resolving it needs individual records.

# 07 The predicted-stay measure in the extract

A measure the files carry, which the application does not use.

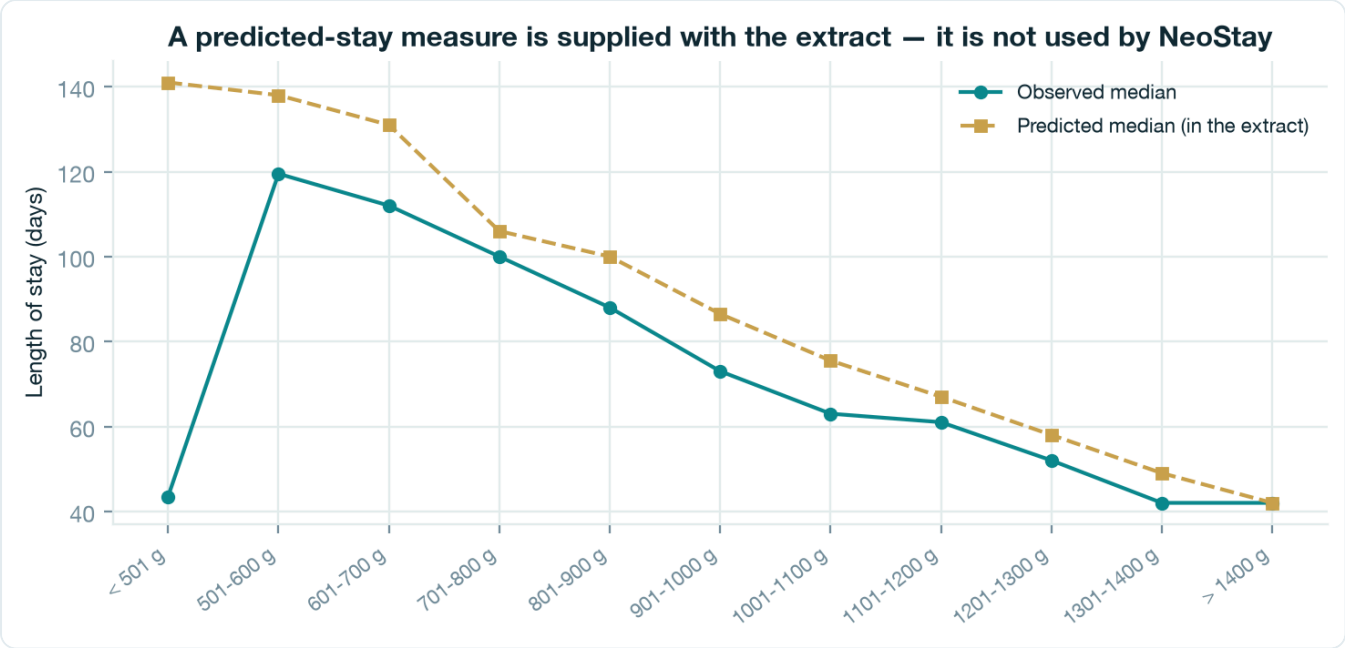



**Figure 4 — A predicted length of stay is supplied with the extract.** It sits above the observed median in every band except the heaviest, and covers 618 infants rather than 712.

Alongside the observed figures, the extract carries a measure named *Predicted Length of Stay*. Its method is not documented in the files. Three observations:

- **It covers a smaller group.** 618 infants against the 712 in the disposition table — and the shortfall is largest in the smallest band (15 of 38).
- **It runs high.** In ten of eleven bands its median exceeds the observed median, by 6 to 19 days in the bands above 501 g.
- **It diverges sharply at the extreme.** In the `< 501 g` band it gives 141 days against an observed 43.5 — consistent with a measure that projects the stay of a surviving infant and therefore excludes early deaths.

**Not used, and deliberately so.** NeoStay builds its own forecast from the observed medians and quartiles. This measure is described here because any reader of the raw files will find it, and because asking the centre how it was produced — and on which infants — is a cheap question with a useful answer: if it is a validated vendor model, it becomes a benchmark to compare against.

# 08 The acuity instrument as data

A policy, not a sample — and what can still be measured about it.

The second dataset is the unit's own acuity policy — the Abrazo Arrowhead Neonatal Acuity Tool and the Nursing Levels of Care — transcribed from a three-page scan into tables and parsed by `backend/app/acuity_tool.py`. The scan's handwritten annotations are deliberately ignored.

> **What cannot be measured.** The instrument carries no counts and no outcomes, so nothing held today shows how often each finding occurs, how the case mix distributes across the levels of care, or whether the prescribed ratios match the nursing effort actually spent. That needs the shift-level acuity record (D5) and actual staffing (D6).

| 11 | 43 | 64 | 108 |
|---|---|---|---|
| BODY SYSTEMS | FINDINGS IN THE GRID | LEVEL-OF-CARE CRITERIA | SELECTABLE ITEMS IN ALL |

Two structural facts follow:

- **Respiratory, nutrition and neurological care carry the most detail** — seven, seven and six findings — where a charge nurse's judgement is most often needed.
- **Four findings trigger the footnoted 1:1 rule:** therapeutic hypothermia for HIE, multiple bili-lights with IV hydration or exchange transfusion, a bedside surgical procedure, and the haemodynamically unstable infant needing intervention every 15 minutes.

TABLE 8 — THE LEVELS OF CARE, BY LEVEL

| Criteria group | N1 | N2 | N3 | N4 | Total |
|---|---|---|---|---|---|
| General criteria | 4 | 13 | 15 | 13 | 45 |
| Hyperbilirubinemia criteria | 6 | 5 | 4 | 4 | 19 |
| **All criteria** | **10** | **18** | **19** | **17** | **64** |

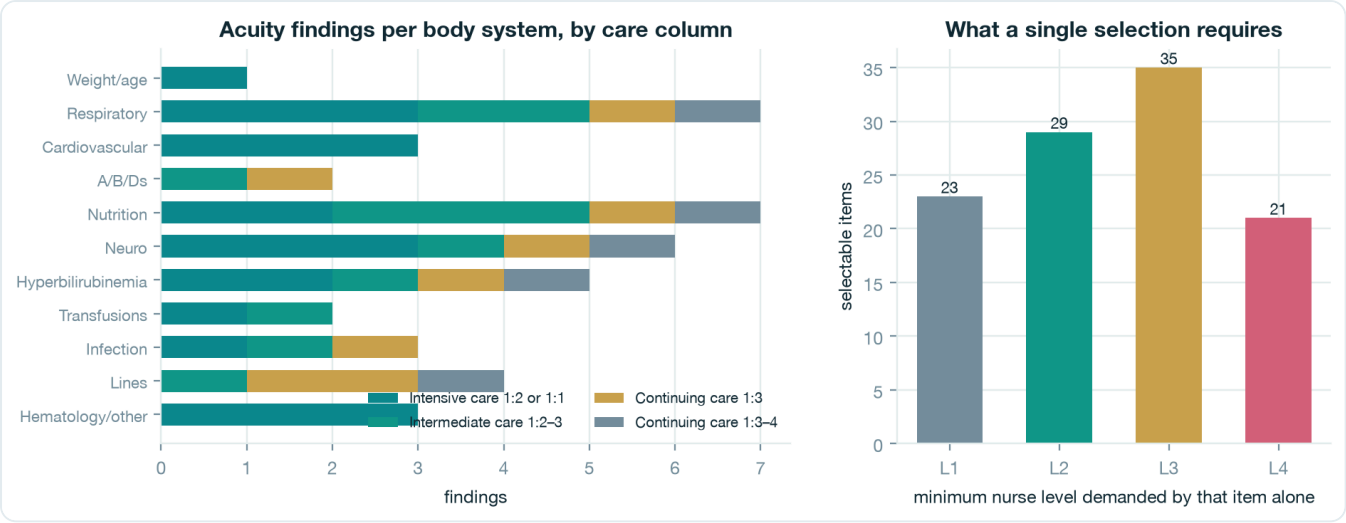



**Figure 5 — The shape of the instrument.** Left: where the clinical detail sits, by body system and care column. Right: the minimum nurse level demanded by each selectable item on its own — 35 of the 108 items require a proficient nurse on their own, and 21 require an expert.

# 09 What these data support, and what they do not

Four of eleven questions can be answered today.

**TABLE 9 — QUESTIONS AGAINST THE DATA HELD TODAY**

| Question | Today | Why |
|---|---|---|
| Expected stay for a group of similar infants | yes | Medians and quartiles per weight band and per gestational week. |
| Chance of home, transfer or death for a group | yes | Counts per band in the disposition table. |
| How much evidence stands behind an estimate | yes | N is published with every statistic, and drives the confidence grade. |
| Nurse:patient ratio and minimum competency per infant | yes | The acuity instrument is complete and executable. |
| A stay prediction for an individual infant | no | Needs patient-level records (D1). No individual rows exist. |
| Effect of a clinical course on the stay | no | Needs the daily course (D2). Weight and gestation are the only covariates held. |
| Joint effect of weight and gestation | no | The two tables describe the same cohort separately; no cross-tabulation is published. |
| Calibration, discrimination, external validation | no | Needs individual outcomes to compare predictions against. |
| Unit demand over time, seasonality, surges | no | Needs census and flow (D4). The files hold no dates. |
| Whether the acuity tool matches real nursing effort | no | Needs acuity per shift (D5) and actual staffing (D6). |
| Rostering real named nurses | no | Needs the workforce list (D6). No staff data is held. |

The pattern is consistent: **what the data supports, the application already does** — group estimates with honest ranges, outcome mix, evidence counts and acuity-driven staffing. Everything in the "no" column waits on an extract from the hospital, not on development work. The six datasets, their fields and the volumes needed are specified in *docs/NeoStay-Data-Requirements.pdf*.

> **The cheapest next step.** Three of the four modelling blockers are resolved by one extract: a pseudonymised row per admission with birth weight, gestational age, admission and discharge dates and outcome. That single file converts group estimates into a fitted model, makes competing-risk analysis possible, and gives the joint weight-and-gestation effect that cannot be recovered from the margins.

# 10 Reproducing this analysis

One notebook, no hidden steps.

Every number, table and figure in this document is produced by `notebooks/01_dataset_analysis.ipynb`, which reads only the files in `datasets/` and writes its figures into `docs/analysis/figures/`.

```
make setup

  .venv/bin/pip install -r notebooks/requirements.txt

  # open notebooks/01_dataset_analysis.ipynb in VS Code, or re-run it from the command line:

  .venv/bin/python -m jupyter nbconvert --to notebook --execute --inplace \

      notebooks/01_dataset_analysis.ipynb
```

TABLE 10 — WHAT THE NOTEBOOK CONTAINS

| Section | Output |
|---|---|
| 1 · Inventory | Every file in `datasets/`, its size, and what reads it. |
| 2 · Reading the tables | An independent parser, cross-checked against the application's loader. |
| 3 · Consistency | The four checks in Table 3, and the denominator reconciliation in Table 4. |
| 4–7 · Outcomes | Figures 1 to 4 with the tables behind them. |
| 8 · Acuity instrument | Figure 5, the structure table, and the classification of every item. |
| 9 · Limits | Table 9, generated rather than written by hand. |
| 10 · Provenance | A SHA-256 digest of every source CSV, so a reader can confirm the files have not changed. |

**Scope of this document.** It describes the data held by the project on 25 October 2026 and nothing else. It makes no clinical claim, and it is not a validation of the application: NeoStay remains a research and decision-support prototype, not a medical device.